

**Part 1 Multiple Choice**  
**30 questions 2 points each.**

1. Which is a valid declaration and initialization of a double?
  - a) `double d = new 2.1;`
  - b) `double d = 2.1f;`
  - c) `double d = "2.1";`
  - d) `double d = 2.1;`
2. What is byte code in the context of Java?
  - a) The type of code executed by a Java compiler.
  - b) `The code executed by a Java Virtual Machine`
  - c) It is another name for a Java source file.
  - d) The type of code generated by a Java Virtual Machine.
3. We wish to create a method named `subtract` that will receive 2 int parameters and return an integer. Which method header is correct?
  - a) `public static void subtract(int a, int b) {`
  - b) `public static int subtract(int a){`
  - c) `public static int subtract(int a, int b) {`
  - d) `public static int subtract(){`
4. We wish to create a method named `checkTrue` that will receive 2 String parameters and compare them, returning a boolean value. Which method header is correct?
  - a) `public static void checkTrue(String a, String b){`
  - b) `public static void checkTrue(String a, String b) ;`
  - c) `public static boolean checkTrue(String a) {`
  - d) `public static boolean checkTrue(String a, String b) {`
5. Look at the method header. What is the correct way to call this method?  
  
`public static double getResult(int var1, double var2){`
  - a) `double out = getResult(2, 2.5);`
  - b) `double out = getResult(2.3,2.1);`
  - c) `int out = getResult(6,1.11);`
  - d) `double out = getResult{11,4.1};`

6. Using the code on the right, what is the value of x?

a) 10  
b) 100  
c) 201  
d) 221

```
int[] anArray = {221, 100, 201, 10, 99, 7};  
  
int x = anArray[2];
```

7. Using the array on the right, what code will return the value 7?

a) `int i = anArray[6];`  
b) `int i = anArray[5];`  
c) `int i = anArray[4];`  
d) `int i = anArray[0];`

```
int[] anArray = {221, 100, 201, 10, 99, 7};
```

8. How many times will this for loop on run?

a) 0 times  
b) 6 times  
c) 7 times  
d) 8 times

```
for(int i = 0; i <= 7; i++){  
  
    System.out.println("Value is " + i);  
  
}
```

9. We want the while loop on the right to continue while the two strings are not equal, or b is true. How would you complete the while loop condition?

a) `while(!(s.equals(t)) || b == false) {`  
b) `while(!(s.equals(t)) && b = true) {`  
c) `while(!(s==t) || b = true) {`  
d) `while(!(s.equals(t)) || b == true) {`

```
String s = "A String";  
String t = "Another String";  
Boolean b = false;  
while(_____) {  
    System.out.println("Display output");  
}
```

10. Look at the code on the right. What technique is being used in the following code snippet?

```
a = (int) (b * c);
```

a) Conversion  
b) Data Structures  
c) Casting  
d) Encapsulation

```
int a = 0;  
double b = 2.5;  
double c = 4;  
int d = 3;
```

```
a = (int) (b * c);  
c++;  
b = a/c;  
d = (int) ((a+3) %b);
```

11. Using the code on the right, after the program has completed, what is the value of a?

- a) 10
- b) 10.0
- c) 0
- d) null

12. Using the code above right, after the program has completed, what is the value of b?

- a) 2.0
- b) 10
- c) 0
- d) 2.5

13. Using the code above right, after the program has completed, what is the value of c?

- a) 5.5
- b) 0
- c) 5.0
- d) 4.0

14. Using the code above right, after the program has completed, what is the value of d?

- a) 11.0
- b) 1.0
- c) 11
- d) 1

```
int a = 0;
double b = 2.5;
double c = 4;
int d = 3;

a = (int) (b * c);
c++;
b = a/c;
d = (int) ((a+3) %b);
```

Look at the following class outline:

```
public class Student extends Person { (Q. 15)

    private int ID;
    private String name;

    public Student(String name, int age (Q. 16)) {
        super(age); (Q.20)
    } (Q.17)

    public int getId(){} (Q.18)

    public void setId(int id){}

    public String toString() {} (Q.19)
}
```

15. What Java principle is being demonstrated by the following line of code?

```
public class Student extends Person {
```

- a) Encapsulation
- b) Method Overriding
- c) Parameter Passing
- d) **Inheritance**

16. What Java principle is being demonstrated by the following line of code?

```
public Student(String name, int age)
```

- a) Encapsulation
- b) **Parameter passing**
- c) Inheritance
- d) Exception Handling

17. What Java principle is being demonstrated by the following line of code?

```
public Student() {
}
```

- a) Exception Handling
- b) **Constructors**
- c) Inheritance
- d) Encapsulation

18. What Java principle is being demonstrated by the following lines of code?

```
private int Id;
```

```
public int getId(){}
```

```
public void setId(int id){}
```

- a) Exception Handling
- b) Parameter Passing
- c) **Encapsulation**
- d) Constructors

19. What Java principle is being demonstrated by the following line of code?

```
public String toString() {}
```

- a) Encapsulation
- b) Parameter Passing
- c) **Method Overriding**
- d) Exception Handling

20. What is happening in the following line of code?

```
super(age);
```

- a) The constructor of the Student object is being called
- b) A new object named age is created
- c) **The constructor of the Person object is being called**
- d) The code is checked for errors

21. Where can myVariable be seen/used?

- a) **Everywhere**
- b) In this package only
- c) In this class only
- d) In the constructor method only

```
public class MyClass{  
    public int myVariable;  
  
    public MyClass(){  
        myVariable = 2;  
    }  
}
```

22. Where can myVariable be seen/used?

- a) In this package only
- b) In the constructor method only
- c) **In this class only**
- d) Everywhere

```
public class MyClass{  
    private int myVariable;  
  
    public MyClass(){  
        myVariable = 2;  
    }  
}
```

23. Where can myVariable be seen/used?

- a) In this package only
- b) In the constructor method only
- c) Everywhere
- d) In the class only

```
public class MyClass{  
    public MyClass(){  
        int myVariable = 2;  
    }  
}
```

Match the concept with the description:

- 24. Overriding (B)
- 25. Inheritance (D)
- 26. Recursion (A)
- 27. Static (E)
- 28. Polymorphism (C)

- A. A method can call itself.
- B. Changing a method inherited from a parent class
- C. Different classes respond in a consistent way to the same method calls.
- D. One class extends another, thereby getting all its properties and behaviour.
- E. Only one version of a class/variable/method exists, and can be used without creating an instance.

29. What method will be called in this code?

- a) go\_nowhere();
- b) fly\_kite();
- c) go\_coffee();
- d) No method called

```
boolean wind_blowing = true;  
int temp = 28;  
int money = 55;  
if (temp > 25) {  
    if(wind_blowing == true) {  
        fly_kite();  
    }  
    else {  
        if (money > 30) {  
            go_coffee();  
        }  
        else {  
            go_cycle();  
        }  
    }  
} else {  
    go_nowhere();  
}
```

30. What is the correct description for the if statement given below?

```
if(temp<=10 && wind_blowing== true && money>100) {  
    go_shopping();  
}
```

- a) If the temperature is less than 10, and the wind is blowing, and money is greater than 100, then call the go\_shopping() method.
- b) If the temperature is less than or equal to 10, and the wind is blowing, and money is greater than 100, then call the go\_shopping() method.
- c) If the temperature is less than or equal to 10, and the wind is not blowing, and money is greater than 100, then call the go\_shopping() method.
- d) If the temperature is equal to 10, and the wind is not blowing, and money is less than 100, then call the go\_shopping() method.

**Part 2 Code Completion: 40 points**

**2 Questions, 20 points each.**

**Question 1**

You have a main class:

```
public class Exam {  
  
    public static void main(String[] args) {  
  
        String[] contents = {"Bottle", "Pen", "Notepad"};  
        String brand = "Adidas";  
        capacity = 25;  
        String colour = "Black"  
  
        Bag schoolbag = new Bag(brand, capacity, colour, contents);  
        System.out.println(schoolbag.toString());  
    }  
}
```

This creates a new Bag object, using the Bag class:

```
public class Bag {  
  
    public static void main(String[] args) {  
  
        // Instance variables  
  
        // Constructor  
  
        // Encapsulation  
  
    }  
}
```

You should complete the Bag class.

- Create suitable instance variables, based on the method call in the Exam class. (5 marks)
- Write a Constructor that will handle the arguments passed in. (5 marks)
- Encapsulation must be used for all instance variable, with getters and setters. (5 marks)
- You should override the toString() method, and create your own. (5 marks)



Question 2: 20 points

Given this Java code:

```
public class Exam

    public static void main(String[] args){
        int[] numbers = new int[10];
        fillArrayRandom(numbers);
        System.out.println("Array contents: ")
        printArray(numbers);
        System.out.println("The sum of the numbers is: "+ findSum(numbers));
        printEvenArray(numbers);
    }

    //fill the array with random integers between 0 and 100
    //use Math.random() which gives a double between 0 and 1

    public static void fillArrayRandom(int[] toFill){
        //missing code (a)
    }

    //print the contents of the array to the console, showing the index and value of
    //each element on a new line

    public static void printArray(int[] toPrint){
        //missing code (b)
    }

    //print the content of every index in the array, but only if the value at that index is an
    // even number. Print each element on a new line

    public static void printEvenArray(int[] toPrint){
        //missing code (c)
    }

    //return the sum of the array
    public static int findSum(int[] array){
        //missing code (d)
    }
}
```

(See next page for further instructions)

Write additional Java code to complete the static methods that are incomplete.  
For each method show the method header and complete the missing code:

- (a) Code the fillArray method to populate the array with random integers between 0 and 100. (5 marks)
- (b) Code the printArray method to print the array contents with each number displayed on a new line. Each array value should be printed on a new line. (5 marks)
- (c) Code the printEvenArray method to print only array contents that are an even number. For example, print the integer 22, but do not print the integer 23. (5 marks)
- (d) Code the findSum method to produce the sum of all the contents of the array. (5 marks)

**END OF FINAL EXAM**